Neural Network

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Introduction

- Neural network are a method in artificial intelligence inspired by the human brain. They fall under a type of machine learning called deep learning. Neural networks use interconnected nodes or neurons in a layered structure. This structure includes an input layer, hidden layers, and an output layer, resembling the way the human brain processes information.
- The learning process involves feeding data into the network and adjusting the connections based on the output error. This is done through a method called backpropagation, which adjusts the weights of the connections to minimize errors.
- Neural network are capable of solving complex problems with high accuracy. They excel in various applications such as image recognition, natural language processing, converting spoken language into text, etc

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- Neural networks have several use cases across many industries. These are four of the important applications of neural networks:

Purpose of neural network

- Computer vision: The ability of computers to extract information and insights from images and videos. With neural networks, computers can distinguish and recognize images similar to humans.
  - Medical diagnosis by medical image classification
  - Facial recognition to identify faces and recognize attributes like open eyes, glasses, and facial hair
  - Image labeling to identify brand logos, clothing, safety gear, and other image details

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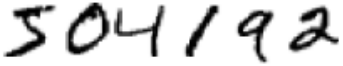
- Recommendation engines: Neural networks can track user activity to develop personalized recommendations. They can also analyze all user behavior and discover new products or services that interest a specific user.

Neuron network mechanism

- The human visual system is incredibly sensitive, with the ability to recognize small details in the surrounding environment
- In fact, the sharpness of our eyesight is thanks to the human brain. The brain contains millions of visual nerve cells, with connections between them to carry out image processing, whether simple or complex. We carry a supercomputer in our heads.



Neural network mechanism

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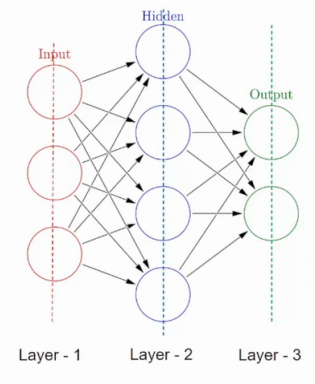
- Output layer: The output layer gives the final result of all the data processing by the artificial neural network.

How do neural network work



Functioning

- Feedforward process: The feedforward process is the initial phase where input data is processed through the network to generate an output. Here's a step-by-step breakdown:
  - Input Layer Reception
  - Propagation Through Hidden Layers
  - Neuron Activation
  - Output Layer Calculation
  - Prediction or Output

Functioning

- Once the network generates an output, the backpropagation process begins. This phase adjusts the network's weights based on the error between the predicted output and the actual output. Here's how it works:
  - Error Calculation
  - Gradient Descent
  - Weight Update
  - Propagation of Error
  - Iterative Learning

Gradient decent

Gradient descent is a crucial method used for optimizing neural networks. In neural networks, its primary objective is to adjust parameters such as weights and biases to minimize the loss function, which measures the discrepancy between the network's predictions and the actual values in the training data. This process begins with computing the gradient of the loss function with respect to each parameter of the neural network using backpropagation.

- Parameter Initialization
- Gradient Calculation
- Parameter Update
- Iteration

Gradient decent: Loss Function

- A loss function, also known as a cost function or objective function, plays a fundamental role in machine learning and neural networks. Its primary purpose is to quantify how well a model's predictions match the actual target values in the training data. The loss function essentially measures the discrepancy between predicted values and ground truth, providing a metric that guides the optimization process during training
- During training, the goal is to minimize the value of the loss function. This process involves adjusting the model parameters (weights and biases in neural networks) iteratively to reduce prediction errors and improve the model's accuracy

Gradient decent: Learning rate

- The learning rate is a crucial parameter in gradient descent methods. It determines the speed and size of adjustments made to model parameters during training
- Impact on Learning Process:
  - A high learning rate can cause large updates in parameter values, potentially overshooting the optimal point or making the learning process unstable (overshooting)
  - A low learning rate slows down the convergence towards the optimal solution, prolonging training time.
- Choosing an appropriate learning rate is often a trial-and-error process, evaluated based on model performance on validation datasets. Techniques such as automatic search and gradual decrease of the learning rate during training can be applied to improve model stability and performance.

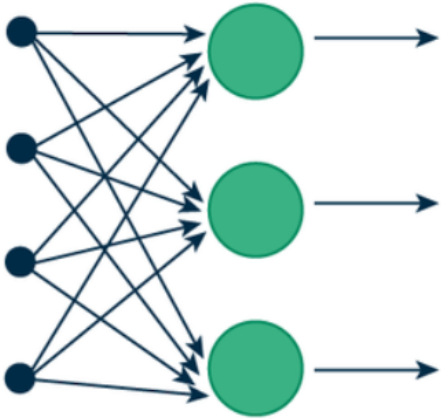
Training

Neural network training is the process of teaching a neural network to perform a task. Neural networks learn by initially processing several large sets of labeled or unlabeled data. By using these examples, they can then process unknown inputs more accurately.

- Supervised learning
- Unsupervised Learning
- Reinforcement Learning

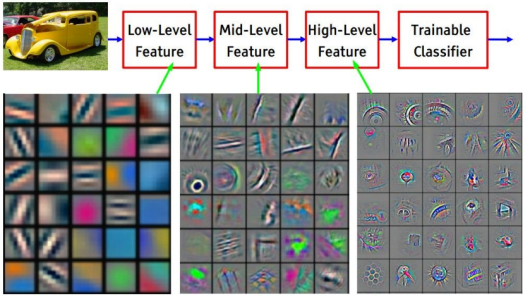
Types of neural network

- Feedforward neural networks: It process data in one direction, from the input node to the output node. Every node in one layer is connected to every node in the next layer. A feedforward network uses a feedback process to improve predictions over time.



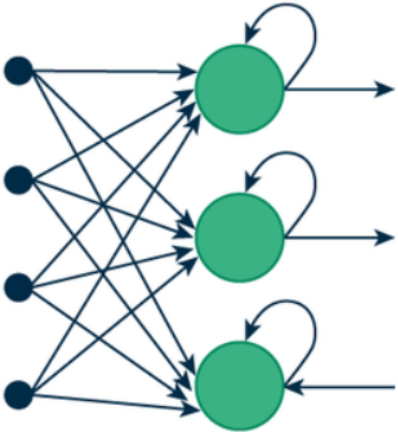
Types of neural network

- Convolutional Neural Network (CNN) is a specialized type of deep learning architecture designed to excel at processing data with a grid-like structure, particularly images and videos.



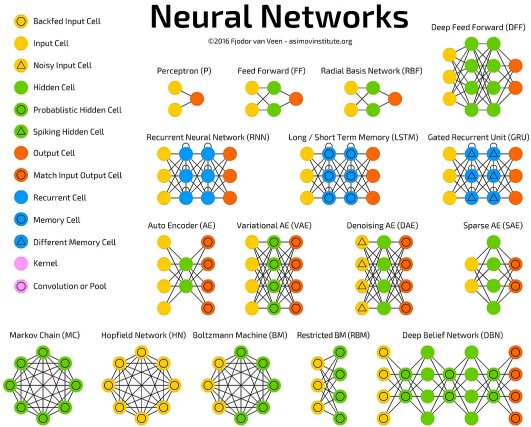
Types of neural network

- Recurrent Neural Network (RNN): are specifically designed to handle sequential data, where the order of information matters. Imagine an FNN trying to predict the next word in a sentence; it wouldn't consider the context of previous words. RNNs address this limitation by incorporating a memory of past information.
- RNNs are used in Language processing, Time series forecasting, Music generation,...



Types of neural network

- Long Short-Term Memory (LSTM) Network, Gated Recurrent Unit (GRU) Network, Transformer Network,...
- There can be many uses of neural network concept tailored to specific purposes. This makes Artificial Neural Network a cornerstone of modern AI development.



Deep learning in context of neural network

- Machine learning is an artificial intelligence technique that gives computers access to very large datasets and teaches them to learn from this data. Machine learning software finds patterns in existing data and applies those patterns to new data to make intelligent decisions.

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- Deep learning is a subset of machine learning that uses deep learning networks to process data.

Deep learning in context of neural network

- Traditional machine learning methods require human input for the machine learning software to work sufficiently well. A data scientist manually determines the set of relevant features that the software must analyze. This limits the software's ability, which makes it tedious to create and manage.
- On the other hand, in deep learning, the data scientist gives only raw data to the software. The deep learning network derives the features by itself and learns more independently. It can analyze unstructured datasets like text documents, identify which data attributes to prioritize, and solve more complex problems.

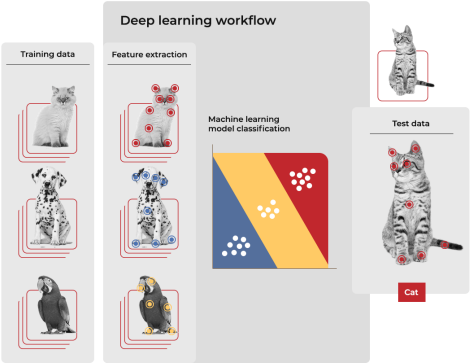
Example for deep learning

If you were training a machine learning software to identify an image of a pet correctly, you would need to take these steps:

- Find and label thousands of pet images, like cats, dogs, horses, hamsters, parrots, and so on, manually.
- Tell the machine learning software what features to look for so it can identify the image using elimination. For instance, it might count the number of legs, then check for eye shape, ear shape, tail, fur, and so on.
- Manually assess and change the labeled datasets to improve the software's accuracy. For example, if your training set has too many pictures of black cats, the software will correctly identify a black cat but not a white one.

Example for deep learning

In deep learning, however, the neural networks would process all the images and automatically determine that they need to analyze the number of legs and the face shape first, then look at the tails last to correctly identify the animal in the image.



Summary

Neural networks are powerful tools in AI, capable of learning from data to perform complex tasks. Their layered structure, ability to learn and adapt, and wide range of applications make them integral to many modern technologies.

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